

TAMIL NADU STATE BOARD - STANDARD 11 CHEMISTRY VOLUME 2

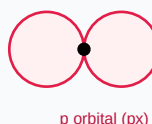
CHAPTER 8: PHYSICAL AND CHEMICAL EQUILIBRIUM

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Physical and Chemical Equilibrium

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

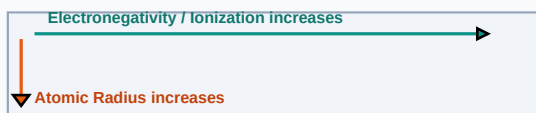
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Reversible Reaction	A reaction that can proceed in both forward and reverse directions, forming equilibrium.
Le Chatelier Principle	If a constraint is applied to a system at equilibrium, the system shifts to counteract that constraint.
Equilibrium Constant	The ratio of the product of concentrations of products to reactants, raised to stoichiometric powers.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

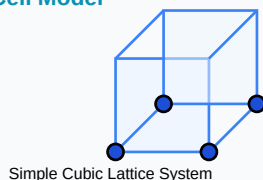
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Kp and Kc relation: $K_p = K_c \cdot (R \cdot T)^{\Delta n_g}$
- Reaction Quotient: $Q = \frac{[\text{Products}]}{[\text{Reactants}]}$ (non-equilibrium states)
- Gibbs relation: $\Delta G_0 = -R \cdot T \cdot \ln(K)$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

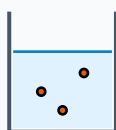
Example 1: Write relation between K_p and K_c for synthesis of Ammonia: $N_2 + 3H_2 \rightleftharpoons 2NH_3$.

Step-by-step Solution:

$\Delta n_g = \text{Moles products} - \text{Moles reactants} = 2 - (1 + 3) = -2$.

$K_p = K_c \cdot (RT)^{-2}$.

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Apply Le Chatelier's principle to evaluate optimal conditions for Ammonia synthesis in Haber's Process.

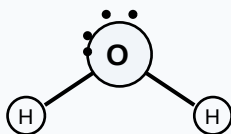
Analytical Solution Strategy:

Reaction: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ (Exothermic, $\Delta H = -92 \text{ kJ}$).

(1) Pressure: 4 moles reactants $\rightarrow$ 2 moles products. Increasing pressure shifts forward.

(2) Temperature: Reaction is exothermic. Lower temperature shifts forward. (Optimal: 450C with Fe catalyst for speed).

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

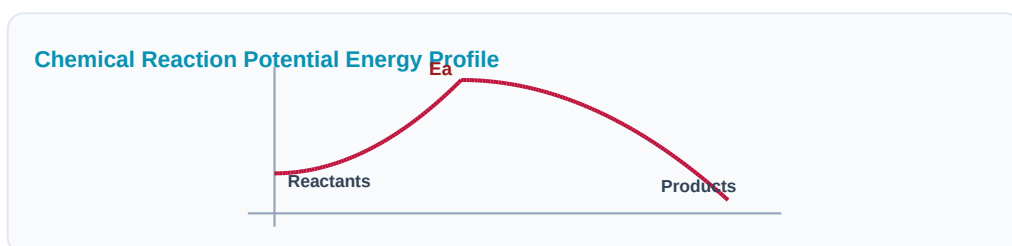
Q1 (1-Mark): State Le Chatelier's Principle.

Q2 (2-Mark): Differentiate between homogeneous and heterogeneous equilibria with examples.

Q3 (2-Mark): If $K_c = 4$ for a reaction at 300K, find K_c for the reverse reaction.

Q4 (5-Mark): Write the expressions of K_p and K_c for thermal decomposition of CaCO_3 .

Q5 (5-Mark): Explain how reaction quotient Q predicts equilibrium shifts.



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Industrial Chemistry

Ammonia yields are maximized by maintaining high pressures of 200 atm.

Application Case: Ocean Acidification

Dissolved carbon dioxide shifts acid carbonate buffers when CO₂ emissions increase.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

